



*Bani sueif national university*

*Faculty of computers and Ai*

# **Banking System Project**

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# INTRODUCTION:

1. **This report presents an overview of the requirements and implementation stages of the Database Systems course project.**
2. **A fully functional database was created, and the tables were properly designed and structured.**
3. **Records were successfully inserted into the database.  
In addition, Structured Query Language (SQL) commands studied during the course were applied.**
4. **A user interface was also developed to facilitate interaction with the database, and it was integrated with the database using the Python programming language.**

# REQUIREMENTS:

1. **ERD & SCHEMA**
2. **Each table contains a constraint**
3. **All the SQL commands that were previously studied.**
4. **GUI contains (insert , update , delete)**

# NOTES:

## i) In ERD :-

- Computed attribute “**NUMBER OF EMPLOYEES**” retrieved by

**SELECT**

**d.name AS department\_name,**

**COUNT(e.employee\_id) AS total\_employees**

**FROM**

**department d**

**LEFT JOIN**

**employee e ON d.department\_code = e.department\_code\_FK**

**GROUP BY**

**d.department\_code, d.name;**

## ii) In schema

- We add 2 tables because the multivalued attribute “**PHONE**” and the relationship many to many “**DEAL WITH**”
- P.K went from 1 to many
- P.K went from partial participation to total participation

# SQL commands:-

## DDL:-

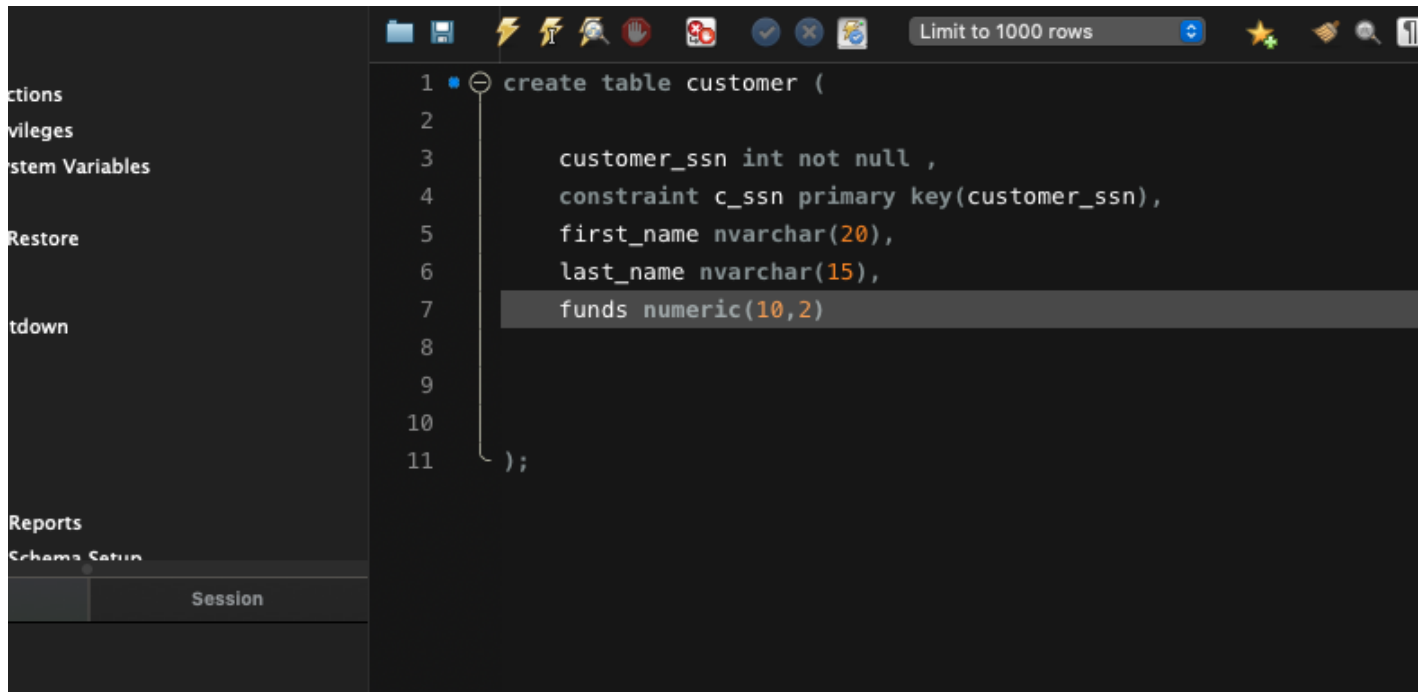
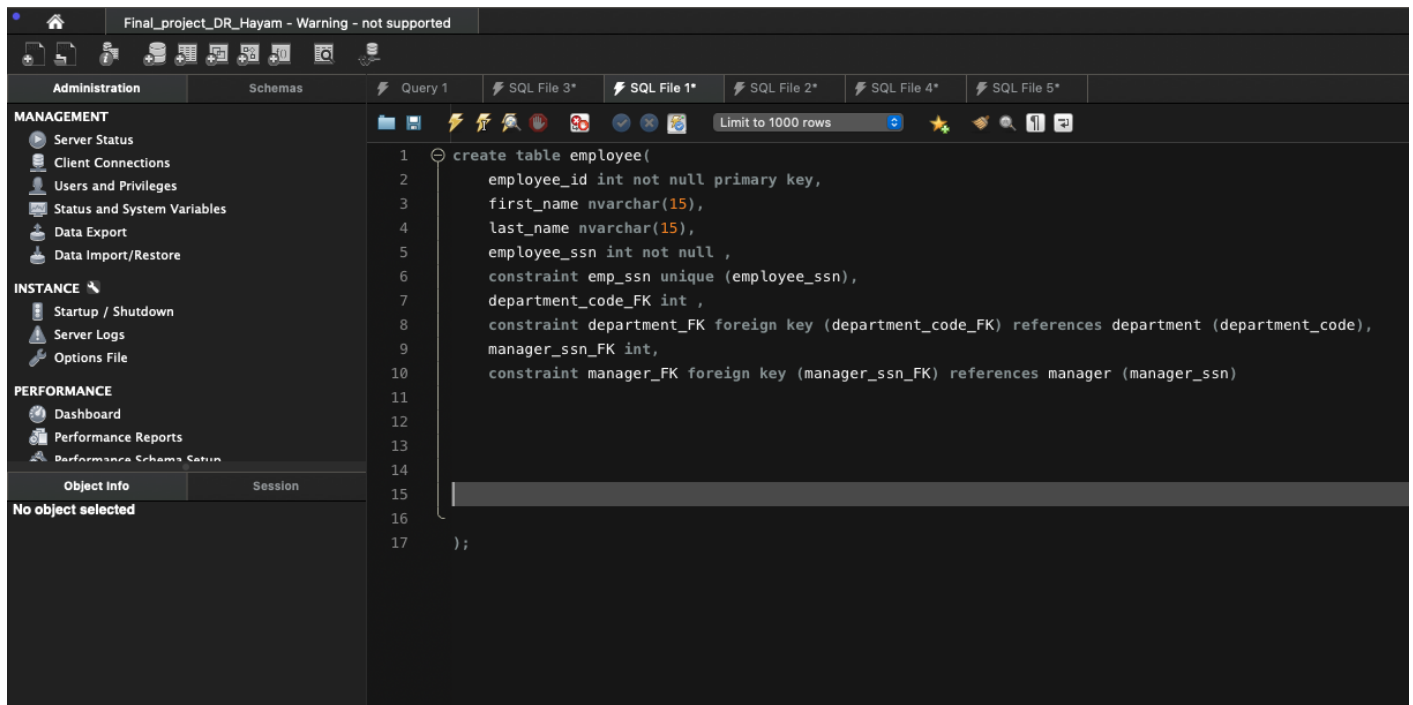
- **CREATE**

The screenshot displays the SQL Enterprise Manager interface with the 'Query 1' tab active. The left-hand navigation pane shows the 'MANAGEMENT' section expanded, with 'Server Status' selected. The main query editor contains the following SQL commands:

```
1 create database Bank;
2 use bank;
3 create table manager(
4     manager_ssn int primary key,
5     name nvarchar(20),
6     branch_name nvarchar(15),
7     salary numeric(10,2),
8     system_access_id int not null,
9     constraint sys_access_id unique (system_access_id)
10 );
```

The second screenshot shows the same interface with the 'Query 1' tab active, displaying the following SQL command:

```
1 create table department(
2
3     department_code int not null,
4     constraint dept_code primary key(department_code),
5     name nvarchar(10)
6
7
8
9
10
11 );
```



Administration Schemas Query 1 SQL File 3\* SQL File 1\* SQL File 2\* SQL File 4\* SQL File 5\*

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Object Info Session

No object selected

```
1 create table deal_with(
2
3     employee_id_FK int ,
4     constraint employee_FK foreign key (employee_id_FK) references employee (employee_id),
5     customer_ssn_FK int,
6     constraint customer_FK foreign key (customer_ssn_FK) references customer (customer_ssn)
7
8
9
10
11
12 );
```

Final\_project\_DR\_Hayam - Warning - not supported

Administration Schemas Query 1 SQL File 3\* SQL File 1\* SQL File 2\* SQL File 4\* SQL File 5\*

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Object Info Session

No object selected

```
1 create table phone(
2
3     phone nvarchar(11) primary key,
4     cus_ssn_FK int ,
5     constraint cus_FK foreign key (cus_ssn_FK) references customer (customer_ssn)
6
7
8
9
10 );
```

- ALTER “ADD”

The screenshot shows the SQL Enterprise Manager interface. On the left, the 'SCHEMAS' tree is expanded to 'Bank' > 'Tables' > 'department' > 'Columns', with 'temp' selected. The main pane shows the query: `alter table department add temp varchar(10);`. The 'Action Output' pane displays the execution results:

|    | Time     | Action                                                                                                                                                                                                                                                                                            |
|----|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | 23:29:24 | create database Bank                                                                                                                                                                                                                                                                              |
| 2  | 23:29:24 | use bank                                                                                                                                                                                                                                                                                          |
| 3  | 23:29:24 | create table manager( manager_ssn int primary key, name nvarchar(20) not null, constraint c_ssn primary key (manager_ssn))                                                                                                                                                                        |
| 4  | 23:30:35 | create table department( department_code int not null, constraint c_dept primary key (department_code))                                                                                                                                                                                           |
| 5  | 23:31:05 | create table employee( employee_id int not null primary key, first_name nvarchar(20) not null, last_name nvarchar(20) not null, email nvarchar(255) not null, phone nvarchar(20) not null, hire_date datetime not null, salary decimal(8,2) not null, constraint c_emp primary key (employee_id)) |
| 6  | 23:31:17 | create table customer ( customer_ssn int not null, constraint c_ssn primary key (customer_ssn))                                                                                                                                                                                                   |
| 7  | 23:31:24 | create table deal_with( employee_id_FK int, constraint employee_id_FK foreign key (employee_id_FK) references employee (employee_id))                                                                                                                                                             |
| 8  | 23:31:32 | create table phone( phone nvarchar(11) primary key, cus_ssn_FK int, constraint cus_ssn_FK foreign key (cus_ssn_FK) references customer (customer_ssn))                                                                                                                                            |
| 9  | 23:36:53 | show tables                                                                                                                                                                                                                                                                                       |
| 10 | 23:43:30 | EXPLAIN alter department add temp varchar(10)                                                                                                                                                                                                                                                     |
| 11 | 23:44:06 | alter table department add temp varchar(10)                                                                                                                                                                                                                                                       |

## MODIFY “ALTER”

The screenshot shows the SQL Enterprise Manager interface. On the left, the 'SCHEMAS' tree is expanded to 'Bank' > 'Tables' > 'department' > 'Columns', with 'temp' selected. The main pane shows the query: `alter table department modify temp int;`. The 'Action Output' pane displays the execution results:

|    | Time     | Action                                                                                                                                                                                                                                                                                            |
|----|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | 23:29:24 | create database Bank                                                                                                                                                                                                                                                                              |
| 2  | 23:29:24 | use bank                                                                                                                                                                                                                                                                                          |
| 3  | 23:29:24 | create table manager( manager_ssn int primary key, name nvarchar(20) not null, constraint c_ssn primary key (manager_ssn))                                                                                                                                                                        |
| 4  | 23:30:35 | create table department( department_code int not null, constraint c_dept primary key (department_code))                                                                                                                                                                                           |
| 5  | 23:31:05 | create table employee( employee_id int not null primary key, first_name nvarchar(20) not null, last_name nvarchar(20) not null, email nvarchar(255) not null, phone nvarchar(20) not null, hire_date datetime not null, salary decimal(8,2) not null, constraint c_emp primary key (employee_id)) |
| 6  | 23:31:17 | create table customer ( customer_ssn int not null, constraint c_ssn primary key (customer_ssn))                                                                                                                                                                                                   |
| 7  | 23:31:24 | create table deal_with( employee_id_FK int, constraint employee_id_FK foreign key (employee_id_FK) references employee (employee_id))                                                                                                                                                             |
| 8  | 23:31:32 | create table phone( phone nvarchar(11) primary key, cus_ssn_FK int, constraint cus_ssn_FK foreign key (cus_ssn_FK) references customer (customer_ssn))                                                                                                                                            |
| 9  | 23:36:53 | show tables                                                                                                                                                                                                                                                                                       |
| 10 | 23:43:30 | EXPLAIN alter department add temp varchar(10)                                                                                                                                                                                                                                                     |
| 11 | 23:44:06 | alter table department add temp varchar(10)                                                                                                                                                                                                                                                       |
| 12 | 23:47:51 | alter table department alter temp int;                                                                                                                                                                                                                                                            |

# DROP

The screenshot displays the SQL Server Enterprise Manager interface. The left pane shows the 'SCHEMAS' tree with the 'Bank' database expanded, showing 'Tables' and 'Columns'. The 'Columns' list for the 'department' table includes 'department\_code', 'name', and 'temp'. The 'Object Info' pane shows the definition of the 'temp' column: 'temp varchar(10)'. The right pane shows the 'Query 1' window with the following SQL script:

```
1 alter table department add temp varchar(10);
2 alter table department modify temp int;
3 alter table department drop column temp;
4
```

The 'Action Output' pane shows the execution results of the script. The final action, 'alter table department drop column temp', is highlighted in blue and marked with a green checkmark.

|      | Time     | Action                                                              |
|------|----------|---------------------------------------------------------------------|
| ✓ 1  | 23:29:24 | create database Bank                                                |
| ✓ 2  | 23:29:24 | use bank                                                            |
| ⚠ 3  | 23:29:24 | create table manager( manager_ssn int primary key, name nvarchar    |
| ⚠ 4  | 23:30:35 | create table department( department_code int not null, constrain    |
| ⚠ 5  | 23:31:05 | create table employee( employee_id int not null primary key, first_ |
| ⚠ 6  | 23:31:17 | create table customer ( customer_ssn int not null , constraint c_s  |
| ✓ 7  | 23:31:24 | create table deal_with( employee_id_FK int , constraint emmploye    |
| ⚠ 8  | 23:31:32 | create table phone( phone nvarchar(11) primary key, cus_ssn_FK      |
| ✓ 9  | 23:36:53 | show tables                                                         |
| ✗ 10 | 23:43:30 | EXPLAIN alter department add temp varchar(10)                       |
| ✓ 11 | 23:44:06 | alter table department add temp varchar(10)                         |
| ✗ 12 | 23:47:51 | alter table department alter temp int:                              |
| ✓ 13 | 23:50:22 | alter table department modify temp int                              |
| ✓ 14 | 23:51:38 | alter table department drop column temp                             |



## • TRUNCATE & DELETE

BEFOOOOOOOOOORE

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Administration Schemas Query 1 SQL File 3\* SQL File 1\* SQL File 2\* SQL File 4\* SQL File 5\* SQL File 6\*

SCHEMAS

Filter objects

- Bank
  - Tables
    - customer
    - deal\_with
    - department
    - employee
    - manager
    - phone
    - temp\_table
      - Columns
        - temp\_pk
        - name
      - Indexes
      - Foreign Keys
      - Triggers
    - Views
    - Stored Procedures
    - Functions
  - Gym
  - market
  - market\_v2
  - sys
  - test\_db
  - uni\_db

Object Info Session

No object selected

```

3      temp_pk int primary key ,
4      name varchar(10)
5
6
7
8
9  );
10  insert into temp_table (temp_pk,name) values (2,'dr hayam');
11
12  select * from temp_table;
13
14

```

100% 1:13

Result Grid Filter Rows: Search Edit: Export/Import:

| temp_pk | name     |
|---------|----------|
| 2       | dr hayam |
| NULL    | NULL     |

temp\_table 1

Action Output

|      | Time     | Action                                                                                                                                            |
|------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| ✓ 1  | 23:29:24 | create database Bank                                                                                                                              |
| ✓ 2  | 23:29:24 | use bank                                                                                                                                          |
| ⚠ 3  | 23:29:24 | create table manager( manager_ssn int primary key, name nvarchar(20), branch_name nvarchar(20))                                                   |
| ⚠ 4  | 23:30:35 | create table department( department_code int not null, constraint dept_code primary key(department_code))                                         |
| ⚠ 5  | 23:31:05 | create table employee( employee_id int not null primary key, first_name nvarchar(15), last_name nvarchar(15))                                     |
| ⚠ 6  | 23:31:17 | create table customer ( customer_ssn int not null, constraint c_ssn primary key(customer_ssn))                                                    |
| ✓ 7  | 23:31:24 | create table deal_with( employee_id_FK int, constraint emmployee_FK foreign key (employee_id_FK) references employee(employee_id))                |
| ⚠ 8  | 23:31:32 | create table phone( phone nvarchar(11) primary key, cus_ssn_FK int, constraint cus_FK foreign key (cus_ssn_FK) references customer(customer_ssn)) |
| ✓ 9  | 23:36:53 | show tables                                                                                                                                       |
| ✗ 10 | 23:43:30 | EXPLAIN alter department add temp varchar(10)                                                                                                     |
| ✓ 11 | 23:44:06 | alter table department add temp varchar(10)                                                                                                       |
| ✗ 12 | 23:47:51 | alter table department alter temp int:                                                                                                            |
| ✓ 13 | 23:50:22 | alter table department modify temp int                                                                                                            |
| ✓ 14 | 23:51:38 | alter table department drop column temp                                                                                                           |
| ✓ 15 | 23:55:39 | create table temp_table( temp_pk int primary key, name varchar(10) )                                                                              |
| ✓ 16 | 00:01:48 | insert into temp_table (temp_pk,name) values (2,'dr hayam')                                                                                       |
| ✓ 17 | 00:03:26 | select * from temp_table LIMIT 0, 1000                                                                                                            |

# AFTEEEEEEEEEEEER

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Administration Schemas Query 1 SQL File 3\* SQL File 1\* SQL File 2\* SQL File 4\* SQL File 5\*

SCHEMAS

Filter objects

- Bank
  - Tables
    - customer
    - deal\_with
    - department
    - employee
    - manager
    - phone
    - temp\_table
      - Columns
        - temp\_pk
        - name
      - Indexes
      - Foreign Keys
      - Triggers
    - Views
    - Stored Procedures
    - Functions
    - Gym
    - market

```

3      temp_pk int primary key ,
4      name varchar(10)
5
6
7
8
9  );
10  insert into temp_table (temp_pk,name) values (2,'dr hayam');
11
12  truncate table temp_table;
13  select * from temp_table;
14

```

100% 1:14

Result Grid Filter Rows: Search Edit: Export/Import:

| temp_pk | name |
|---------|------|
| NULL    | NULL |

temp\_table 2

Final\_project\_DR\_Hayam - Warning - not supported

Administration Schemas Query 1 SQL File 3\* SQL File 1\* SQL File 2\* SQL File 4\* SQL File 5\* SQL File 6\* SQL File 7\*

SCHEMAS

Filter objects

- Bank
  - Tables
    - customer
    - deal\_with
    - department
    - employee
    - manager
    - phone
    - Views
    - Stored Procedures
    - Functions
    - Gym
    - market
    - market\_v2
    - sys
    - test\_db
    - uni\_db

```

1  create table temp_table(
2
3      temp_pk int primary key ,
4      name varchar(10)
5
6
7
8
9  );
10  insert into temp_table (temp_pk,name) values (2,'dr hayam');
11  select * from temp_table;
12
13
14  truncate table temp_table;
15  select * from temp_table;
16
17  drop table temp_table;
18  select * from temp_table;
19
20
21

```

100% 1:21

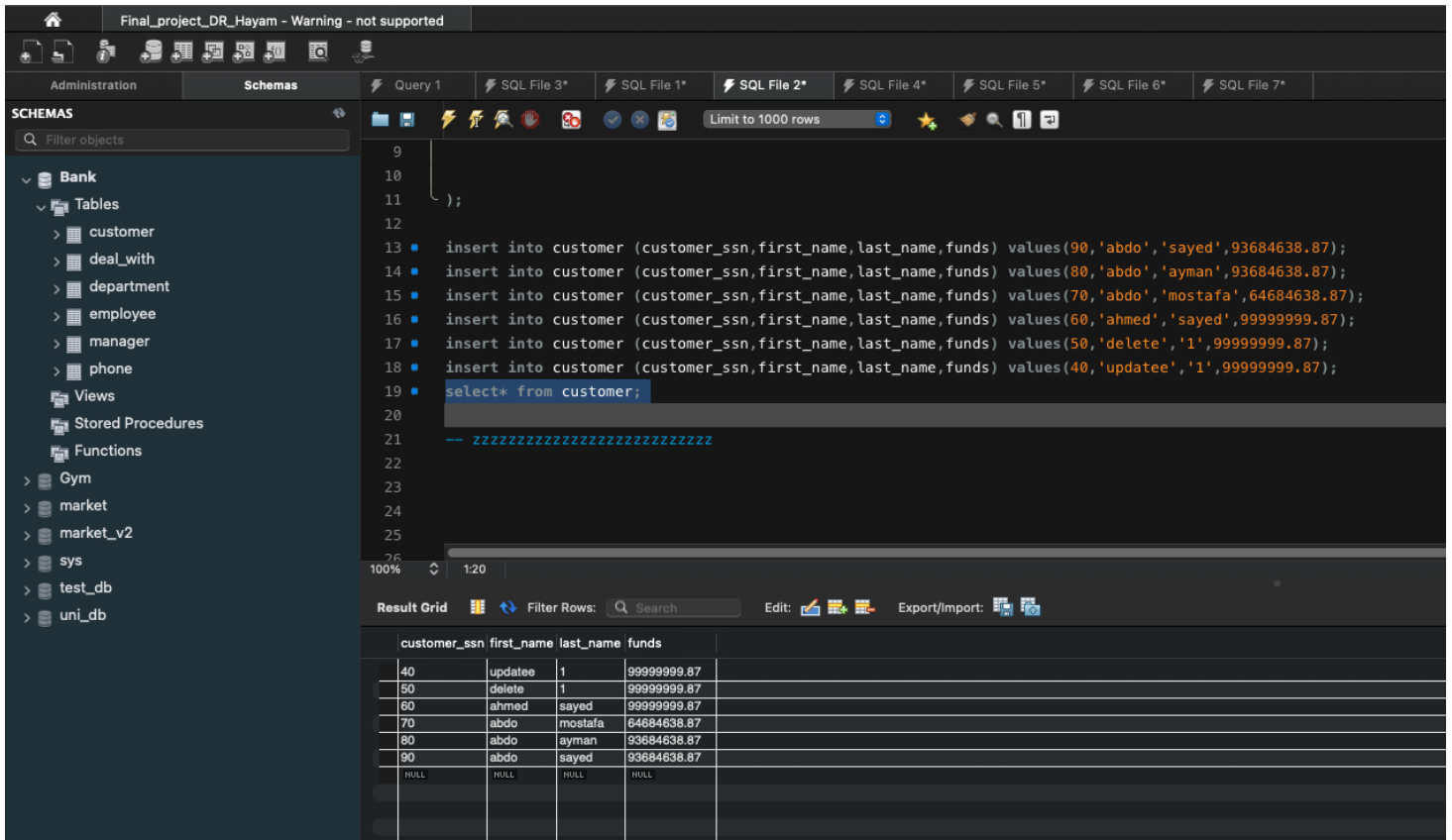
Action Output

- **DML:- INSERT**

[illegible]

- **UPDATE**

# BEFORE

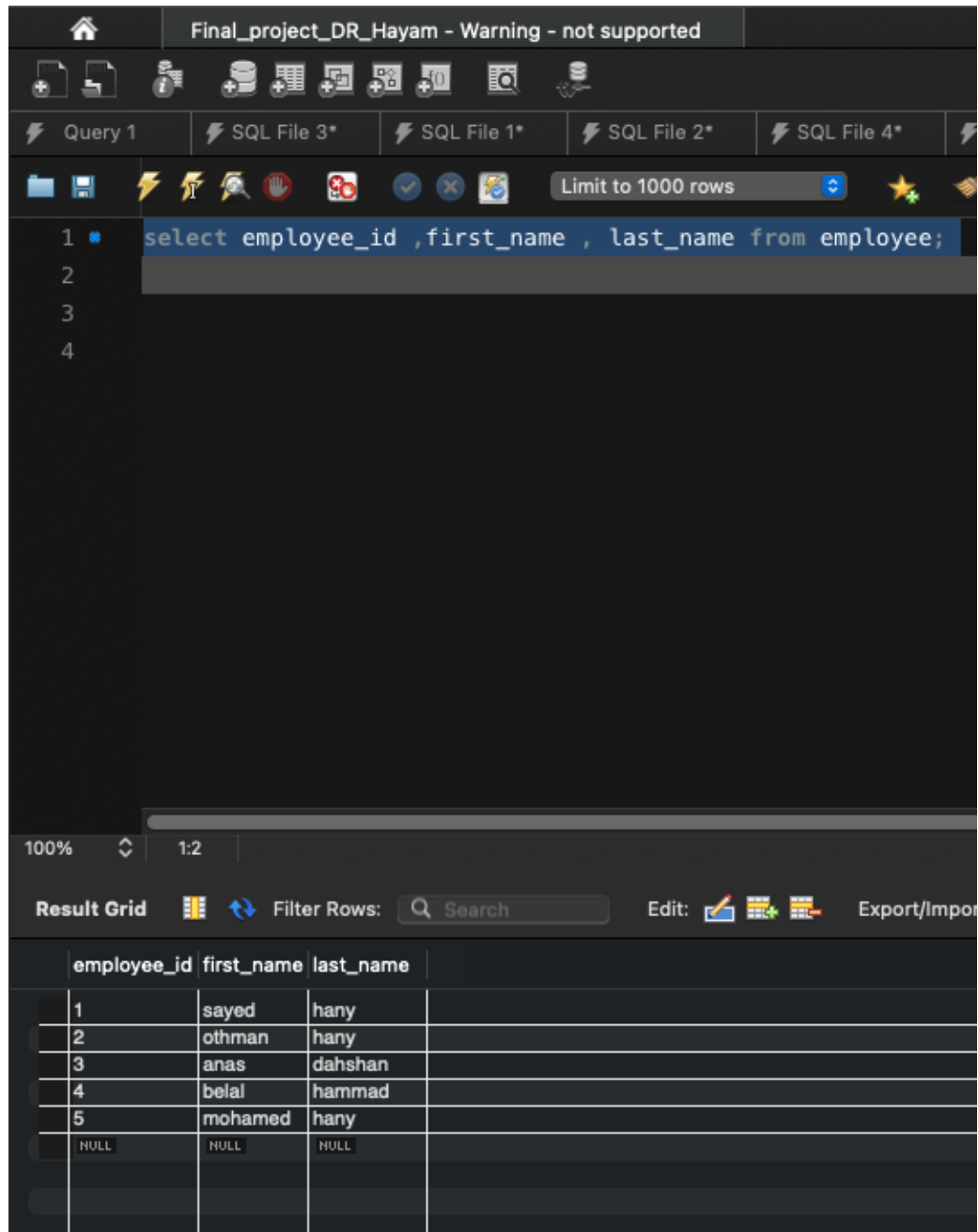


# AFFFFFFFFTER



# SELECT

- **BASIC**



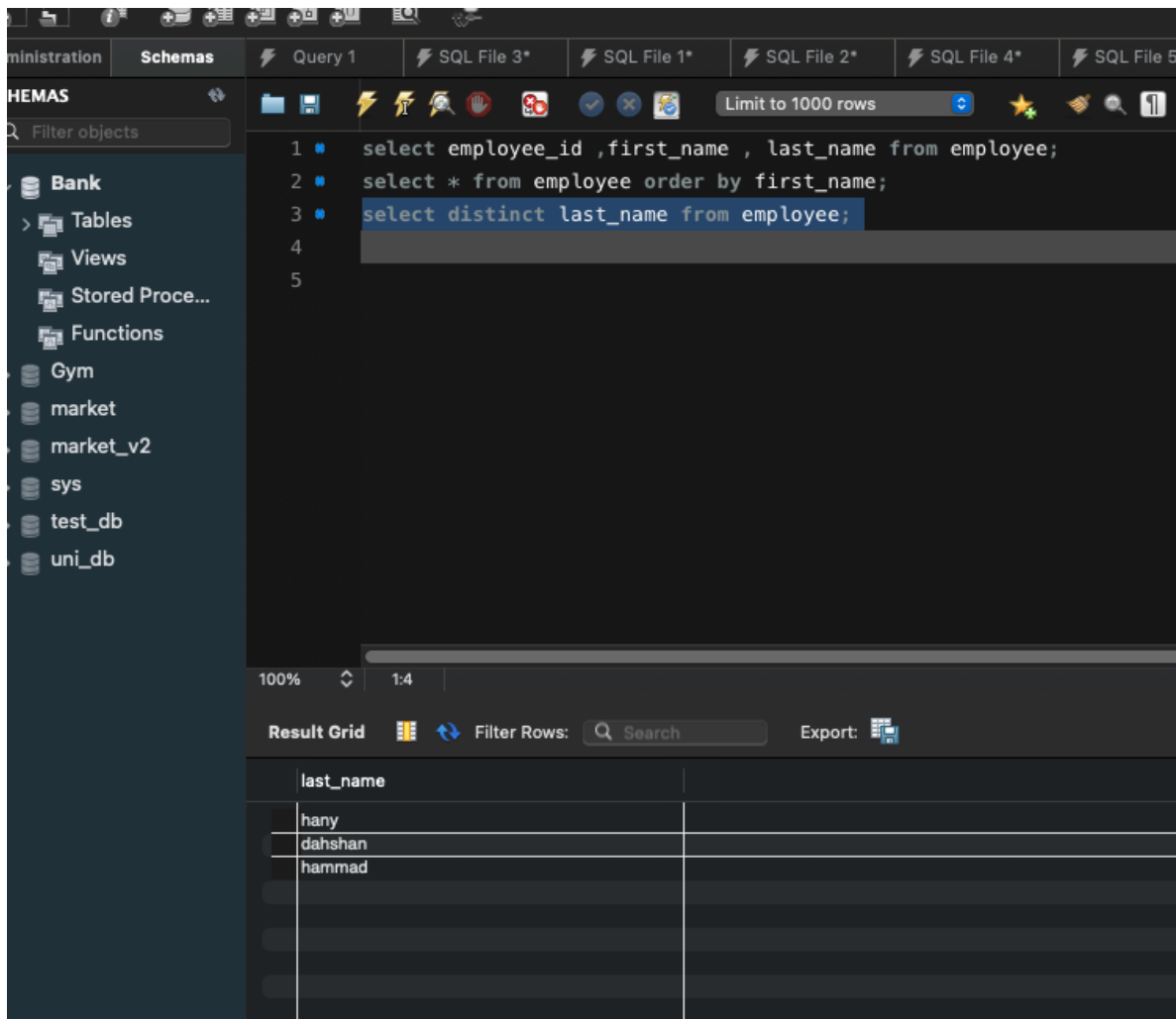
The screenshot shows a SQL IDE interface. The top toolbar includes icons for file operations and a warning message: "Final\_project\_DR\_Hayam - Warning - not supported". Below the toolbar, there are tabs for "Query 1", "SQL File 3\*", "SQL File 1\*", "SQL File 2\*", and "SQL File 4\*". A toolbar below the tabs contains icons for running, saving, and other actions, along with a "Limit to 1000 rows" dropdown. The main editor area displays the following SQL query:

```
1 select employee_id ,first_name , last_name from employee;  
2  
3  
4
```

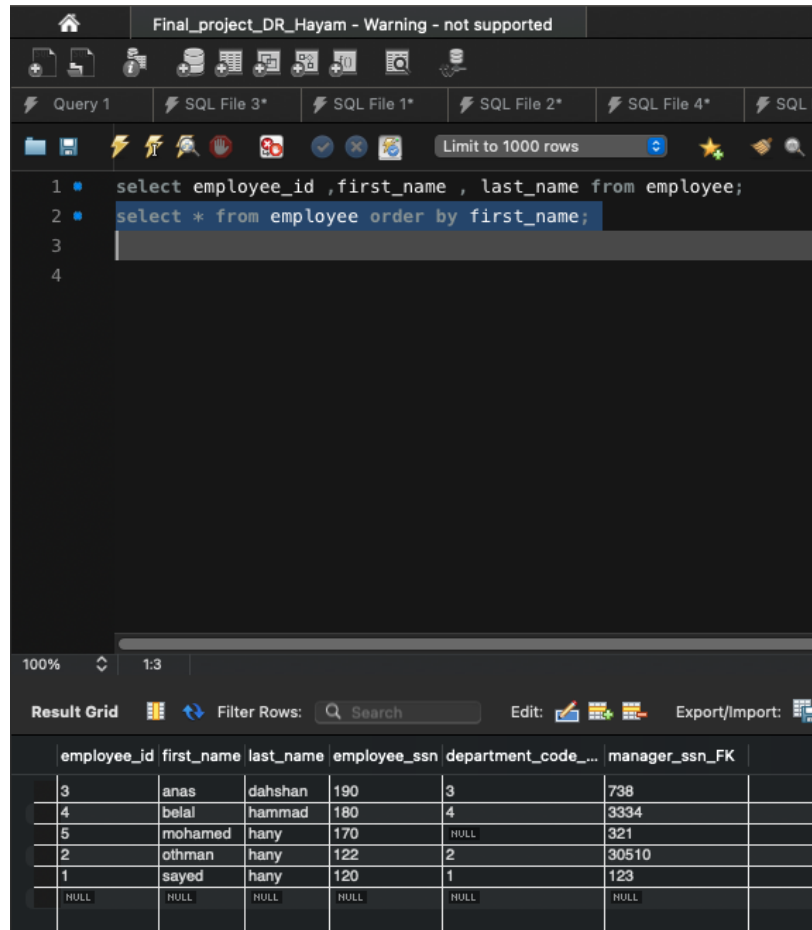
Below the editor, the "Result Grid" is visible, showing the results of the query. The grid has columns for "employee\_id", "first\_name", and "last\_name". The results are as follows:

| employee_id | first_name | last_name |
|-------------|------------|-----------|
| 1           | sayed      | hany      |
| 2           | othman     | hany      |
| 3           | anas       | dahshan   |
| 4           | belal      | hammad    |
| 5           | mohamed    | hany      |
| NULL        | NULL       | NULL      |

- **DISTINCT**



- **ORDER BY**



- **WHERE**  
**LIKE**



Final\_project\_DR\_Hayam - Warning - not supported

Administration Schemas Query 1 SQL File 3\* SQL File 1\* SQL File 2\* SQL File 4\* SQL File 5\* SQL File 6\*

SCHEMAS

Filter objects

- Bank
  - Tables
  - Views
  - Stored Proce...
  - Functions
- Gym
- market
- market\_v2
- sys
- test\_db
- uni\_db

1 select employee\_id ,first\_name , last\_name from employee;

2 select \* from employee order by first\_name;

3 select distinct last\_name from employee;

4 select \* from employee where employee\_ssn between 100 and 172;

5 select \* from employee where first\_name like 's%';

6

7

100% 1:6

Result Grid Filter Rows: Search Edit: Export/Import:

| employee_id | first_name | last_name | employee_ssn | department_code_... | manager_ssn_FK |
|-------------|------------|-----------|--------------|---------------------|----------------|
| 1           | sayed      | hany      | 120          | 1                   | 123            |
|             |            |           |              |                     |                |
|             |            |           |              |                     |                |
|             |            |           |              |                     |                |
|             |            |           |              |                     |                |
|             |            |           |              |                     |                |

Final\_project\_DR\_Hayam - Warning - not supported

Administration Schemas Query 1 SQL File 3\* SQL File 1\* SQL File 2\* SQL File 4\* SQL File 5\* SQL File 6\*

SCHEMAS

Filter objects

- Bank
  - Tables
  - Views
  - Stored Proce...
  - Functions
- Gym
- market
- market\_v2
- sys
- test\_db
- uni\_db

1 select employee\_id ,first\_name , last\_name from employee;

2 select \* from employee order by first\_name;

3 select distinct last\_name from employee;

4 select \* from employee where employee\_ssn between 100 and 172;

5 select \* from employee where first\_name like 's%';

6 select \* from employee where last\_name like '%h%';

7

8

9

100% 1:7

Result Grid Filter Rows: Search Edit: Export/Import:

| employee_id | first_name | last_name | employee_ssn | department_code_... | manager_ssn_FK |
|-------------|------------|-----------|--------------|---------------------|----------------|
| 1           | sayed      | hany      | 120          | 1                   | 123            |
| 2           | othman     | hany      | 122          | 2                   | 30510          |
| 3           | anas       | dahshan   | 190          | 3                   | 798            |
| 4           | belal      | hammad    | 180          | 4                   | 3334           |
| 5           | mohamed    | hany      | 170          | NULL                | 321            |
|             |            |           |              |                     |                |
|             |            |           |              |                     |                |
|             |            |           |              |                     |                |
|             |            |           |              |                     |                |

# BETWEEN

The screenshot shows a database management interface with a dark theme. The top bar indicates the database is 'Final\_project\_DR\_Hayam' with a warning 'Warning - not supported'. The left sidebar shows a tree view of schemas, including 'Bank', 'Gym', 'market', 'market\_v2', 'sys', 'test\_db', and 'uni\_db'. The main area displays a SQL query in a text editor:

```
1 select employee_id ,first_name , last_name from employee;  
2 select * from employee order by first_name;  
3 select distinct last_name from employee;  
4 select * from employee where employee_ssn between 100 and 172;  
5  
6
```

Below the query editor, the 'Result Grid' shows the results of the query. The grid has columns: employee\_id, first\_name, last\_name, employee\_ssn, department\_code\_..., and manager\_ssn\_FK. The results are as follows:

| employee_id | first_name | last_name | employee_ssn | department_code_... | manager_ssn_FK |
|-------------|------------|-----------|--------------|---------------------|----------------|
| 1           | sayed      | hany      | 120          | 1                   | 123            |
| 2           | othman     | hany      | 122          | 2                   | 30510          |
| 5           | mohamed    | hany      | 170          | NULL                | 321            |
| NULL        | NULL       | NULL      | NULL         | NULL                | NULL           |

- JOIN  
INNER

Final\_project\_DR\_Hayam - Warning - not supported

Administration Schemas Query 1 SQL File 3\* SQL File 1\* SQL File 2\* SQL File 4\* SQL File 5\*

SCHEMAS Filter objects

- Bank
  - Tables
  - Views
  - Stored Proce...
  - Functions
- Gym
- market
- market\_v2
- sys
- test\_db
- uni\_db

```

1 select employee_id ,first_name , last_name from employee;
2 select * from employee order by first_name;
3 select distinct last_name from employee;
4 select * from employee where employee_ssn between 100 and 172;
5 select * from employee where first_name like 's%';
6 select * from employee where last_name like '%h%';
7
8
9 -- .....ya rab full mark .....
10
11 select employee_id , first_name,last_name , name from employee inner
12
13
14

```

100% 116:11

Result Grid Filter Rows: Search Export:

| employee_id | first_name | last_name | name       |
|-------------|------------|-----------|------------|
| 1           | sayed      | hany      | sales      |
| 2           | othman     | hany      | CallCenter |
| 3           | anas       | dahshan   | markting   |
| 4           | belal      | hammad    | services   |

LEFT

Final\_project\_DR\_Hayam - Warning - not supported

Administration Schemas Query 1 SQL File 3\* SQL File 1\* SQL File 2\* SQL File 4\* SQL File 5\* SQL File 6\* SQL File 7\*

SCHEMAS Filter objects

- Bank
  - Tables
  - Views
  - Stored Proce...
  - Functions
- Gym
- market
- market\_v2
- sys
- test\_db
- uni\_db

```

1 select employee_id ,first_name , last_name from employee;
2 select * from employee order by first_name;
3 select distinct last_name from employee;
4 select * from employee where employee_ssn between 100 and 172;
5 select * from employee where first_name like 's%';
6 select * from employee where last_name like '%h%';
7
8
9 -- .....ya rab full mark .....
10
11 select employee_id , first_name,last_name , name from employee inner join department on department_code_FK= department_code;
12
13 select employee_id,first_name , last_name , name from employee left outer join department on department_code_FK= department_code;
14
15

```

100% 1:14

Result Grid Filter Rows: Search Export:

| employee_id | first_name | last_name | name       |
|-------------|------------|-----------|------------|
| 1           | sayed      | hany      | sales      |
| 2           | othman     | hany      | CallCenter |
| 3           | anas       | dahshan   | markting   |
| 4           | belal      | hammad    | services   |
| 5           | mohamed    | hany      | HALL       |

## • SUBQUERY & AGG FUNC

The screenshot shows a SQL IDE interface with a query editor and a result grid. The query editor contains the following SQL code:

```
13 select employee_id,first_name , last_name , name from employee left outer join department on department_code_FK= department_code;
14
15 select c.first_name as cus_name , p.phone as cus_phone from customer c right join phone p on p.cus_ssn_FK=customer_ssn;
16 -- full join by ai cause i use mysql
17 SELECT e.first_name, d.name AS department_name
18 FROM employee e
19 LEFT JOIN department d ON e.department_code_FK = d.department_code
20 UNION
21 SELECT e.first_name, d.name AS department_name
22 FROM employee e
23 RIGHT JOIN department d ON e.department_code_FK = d.department_code;
24 -- ya rab fuul mark
25
26 select system__access_id,name, branch_name from manager where salary > (select avg(salary)from manager);
27
28
```

The result grid shows the following data:

| system__access... | name | branch_name |
|-------------------|------|-------------|
| 2233              | abdo | cairo       |

## • VIEW

The screenshot shows a SQL IDE interface with a query editor and a result grid. The query editor contains the following SQL code:

```
19 LEFT JOIN department d ON e.department_code_FK = d.department_code
20 UNION
21 SELECT e.first_name, d.name AS department_name
22 FROM employee e
23 RIGHT JOIN department d ON e.department_code_FK = d.department_code;
24 -- ya rab fuul mark
25
26 select system__access_id,name, branch_name from manager where salary > (select avg(salary)from manager);
27
28 -- viewwwwww
29
30 create view eemp_dept as select first_name , last_name , name from employee join department on department_code_FK= department_code;
31
32 select * from eemp_dept ;
33
34
```

The result grid shows the following data:

| first_name | last_name | name       |
|------------|-----------|------------|
| sayed      | hany      | Sales      |
| olhman     | hany      | CallCenter |
| anas       | dahshan   | marketing  |
| belal      | hammad    | services   |